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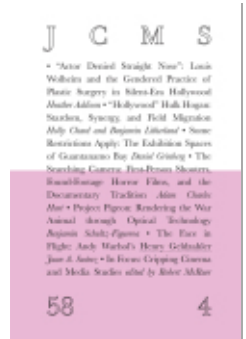
Project Pigeon: Rendering the War Animal through Optical Technology

Benjamin Schultz-Figueroa

JCMS: Journal of Cinema and Media Studies, Volume 58, Number 4, Summer 2019, pp. 92-111 (Article)

Published by University of Texas Press

DOI: <https://doi.org/10.1353/cj.2019.0059>



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Project Pigeon: Rendering the War Animal through Optical Technology

by BENJAMIN SCHULTZ-FIGUEROA

Abstract: This article analyzes the multiple uses of film in B. F. Skinner's classified World War II research into pigeon-guided missiles, arguing that they illustrate an ongoing transformation in the deployment of autonomous nonhuman vision on the battlefield. It concludes that the Project Pigeon films worked to both reimagine and reshape what a combatant could be, no longer dependent on a decision-making human actor but on a complex array of visual interactions between a hybrid organism or device and its environment.

In his 1979 autobiography, *The Shaping of a Behaviorist*, B. F. Skinner recounted a fateful train ride to Chicago in 1940, just after the Nazis had invaded Denmark. Gazing out the train window, the renowned behaviorist ruminated on the destructive power of aerial warfare when his eye unexpectedly caught sight of a “flock of birds lifting and wheeling in formation as they flew alongside the train.”¹ Skinner recounts: “Suddenly I saw them as ‘devices’ with excellent vision and extraordinary maneuverability. Could they not guide a missile?”² Observing the coordination of the flock, its “lifting and wheeling,” inspired in Skinner a new vision of aerial warfare, one that yoked the senses and movements of living animals to the destructive power of modern ballistics. This inspiration began a three-year project to weaponize pigeons, code-named Project Pigeon (Figure 1), by having them guide the flight of a bomb from inside its nose, a project that tied together laboratory research, military technology, and private industry.

This strange story is discussed in popular media as a historical fluke of sorts, a wacky one-off in military research and development. As Skinner himself described, one of the main obstacles to Project Pigeon even at the time was the perception of a pigeon-guided missile as an absurd “crackpot idea.”³ In this article, I argue that

1 B. F. Skinner, *The Shaping of a Behaviorist: Part Two of an Autobiography* (New York: Knopf, 1979), 241.

2 Skinner, 241.

3 B. F. Skinner, “Pigeons in a Pelican,” *American Psychologist* 15, no. 1 (January 1960): 28–37.

Benjamin Schultz-Figueroa is an assistant professor of film studies at Seattle University. He earned his PhD in film and digital media from the University of California, Santa Cruz, and his MA in media studies from the New School. His research focuses on the history of scientific filmmaking, nontheatrical film, and animal representations on film.

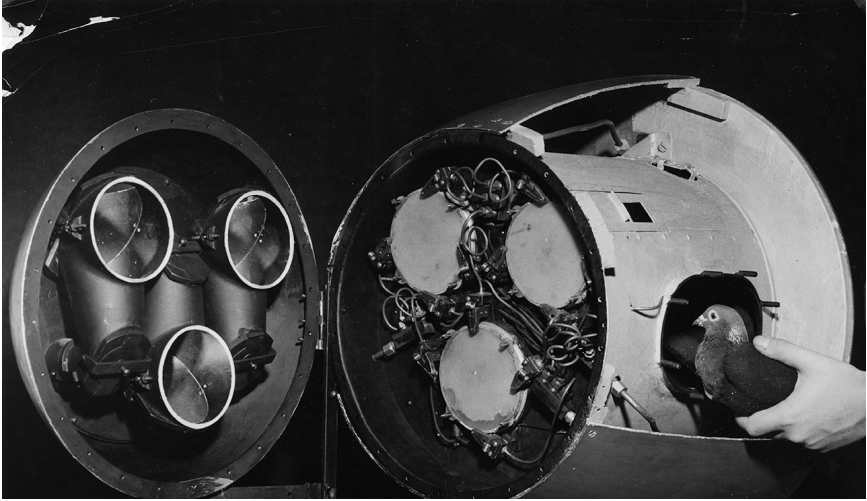


Figure 1. Photograph of Project Pigeon from the Burrhus Frederic Skinner Papers at Harvard University Archives. Courtesy of the B. F. Skinner Foundation.

it is in fact a telling example of the long-standing weaponization of animals, paying special attention to the ways film was used to train pigeons to act as weapons. Historically, Project Pigeon occurred during a crucial military shift away from the model of an elaborate chess game played out by generals and their armies and toward an ecological framework in which a wide array of nonhuman agents play roles. Jussi Parikka has noted a similar shift in artificial intelligence, a movement toward “agents that expressed complex behavior, not through preprogramming and centralization, but through autonomy, emergence, and distributed functioning.”⁴ The missile developed by the Project Pigeon research team was premised on a conversion of the pigeon from an autonomous consciousness to a living machine. Emptied of intentionality, the bird would become only a controllable, yet dynamic and complex, behavior that could be designed and trusted to operate without the oversight of a human commander. Herein we see a reimagining of what a combatant entity can be, dependent no longer on a decision-making human actor but on a complex array of interactions among organism, weapon, and environment. As we will see, the vision of a pigeon-guided bomb presaged the nonhuman sight of the smart bomb, drone, and military robot, where artificial intelligence and computer algorithms replace the operations of animal and human.

Film played a key role in Skinner’s experiment. Media and cinema scholars have written extensively about the visual transformation of the battlefield and film’s place in this history. For instance, militaries from across the globe have used film in dramatically unorthodox ways. Lee Grieveson and Haidee Wasson argue that the US military historically used film as “an iterative apparatus with multiple capacities and

4 Indeed, the information revolution that Parikka describes was closely connected to the studies of distributed control that Skinner’s later work would focus on. Jussi Parikka, *Insect Media: An Archaeology of Animals and Technology* (Minneapolis: University of Minnesota Press, 2010), xi.

functions,” experimenting with the design of the camera, projector, and screen to fit new strategic interests as they arose.⁵ As propaganda, film was used to portray the military to civilians at home and abroad; as a training tool, it was used to consistently instruct large numbers of recruits; and as industrial or advertising aid, it was used by different branches of the military to create reports and communiqués. Project Pigeon added a radically unorthodox use of film to these examples of military innovation in film form. By intervening in the long-standing relationship between the moving image and its audience, the researchers at Project Pigeon marshaled film’s influence on nonhuman viewers as well as humans. As Wasson argues in her chapter dedicated to experimental projection practices, the US Army “boldly dissembled [*sic*] cinema’s settled routines and structures, rearticulating film projection as but one integral element of a growing institution with highly complex needs.”⁶ In the case that follows, I show how military scientists rearticulated film by linking it to other optical media—including camera obscuras and large-format photography—as part of a complex assemblage to control the senses and behaviors of the pigeons. In this assemblage, film becomes enmeshed in a complex intermedia sphere that includes human and nonhuman bodies and sensations. In this hitherto-unstudied use of film by the military, the various components of the moving image were repurposed to transform animals into weapons and combatants.

Rendering Project Pigeon. Project Pigeon was one of the earliest projects to come out of B. F. Skinner’s illustrious and influential career as a behavioral psychologist. After the war, Skinner was one of the best-known voices in American psychology, introducing the “Skinner box” to the study of animal behavior and the vastly influential theory of “operant conditioning.”⁷ His influence was not limited to the sciences but broadly felt across conversations in political theory, linguistics, and philosophy as well. As James Capshew has shown, much of Skinner’s later, more well-known research originated in this military research into pigeon-guided ballistics.⁸ Growing out of initial independent trials in 1940, Project Pigeon secured funding from the US Army’s Office of Scientific Research and Development in 1943.⁹ The culmination of this work placed three pigeons in the head of a missile; the birds had been trained to peck a camera obscura screen showing incoming targets. These pecks were then translated into instructions for the missile’s guidance system. The goal was a 1940s version of a smart bomb, which was capable of correcting course midflight in response to the movement of a target. Although Project Pigeon developed relatively rapidly, the US Army ultimately denied it further funds in December 1943, effectively ending Skinner’s brief oversight of the project. However, in 1948, the US Naval Research

5 Lee Grieveson and Haidee Wasson, “The Military’s Cinema Complex,” in *Cinema’s Military Industrial Complex* (Berkeley: University of California Press, 2018), 7.

6 Haidee Wasson, “Experimental Viewing Protocols: Film Projection and the American Military,” in *Cinema’s Military Industrial Complex* (Berkeley: University of California Press, 2018), 25.

7 B. F. Skinner, *Science and Human Behavior* (New York: Free Press, 1965), 65–66.

8 James H. Capshew, “Engineering Behavior: Project Pigeon, World War II, and the Conditioning of B. F. Skinner,” *Technology and Culture* 34, no. 4 (October 1, 1993): 835–857.

9 Skinner, “Pigeons in a Pelican.”

Laboratory picked up Skinner's research and renamed it Project ORCON—a contraction of “organic” and “control.” Here, with Skinner's consultation, the pigeons' tracking capacity for guiding missiles to their intended targets was methodically tested, demonstrating wide variance in reliability.¹⁰ In the end, the pigeons' performance and accuracy relied on so many uncontrollable factors that Project ORCON, like Project Pigeon before it, was again discontinued. Although the project was tested extensively in the laboratory, it was never used in the field.

Moving images played two central roles in Project Pigeon. First, they provided a means of orienting the pigeons in space and testing the accuracy of their responses, examples of what Harun Farocki calls “operational images.”¹¹ This use of moving-image technology shows up in the final design of Project Pigeon, where each of the three pigeons was constantly responding to camera obscuras installed in the front of the bomb. The pigeons were trained to pinpoint the shape of incoming targets on individual screens (or “plates”) by pecking them as the bomb dropped, which would then cause the bomb to change course. This screen was connected to the bomb's guidance through four small rubber pneumatic tubes that were attached to each side of the frame, which directed a constant airflow to a pneumatic pickup system that controlled the thrusters of the bomb. As Skinner explained: “When the missile was on target, the pigeon pecked the center of the plate, all valves admitted equal amounts of air, and the tambours remained in neutral positions. But if the image moved as little as a quarter of an inch off-center, corresponding to a very small angular displacement of the target, more air was admitted by the valves on one side, and the resulting displacement of the tambours sent appropriate correcting orders directly to the servosystem.”¹² In the later iteration of Project ORCON, the pigeons were tested and trained with color films taken from footage recorded on a jet making diving runs on a destroyer and a freighter, and the pneumatic relays between the servosystem and the screen were replaced with electric currents. Here, the camera obscura and the training films were used to integrate the living behavior of the pigeon into the mechanism of the bomb itself and to produce immersive simulations for these nonhuman pilots in order to fully operationalize their behavior.

The second use of moving images for this research was as a tool for convincing potential sponsors of the pigeon's capacity to act as a weapon. This initiative was realized in a set of promotional films for Project Pigeon, which Skinner largely credited for procuring its initial funding from General Mills Inc. and later the US Navy's renewal of the research as Project ORCON. Skinner's letters indicate that there were multiple films made for this purpose, which were often recut to incorporate new footage.¹³ Skinner's promotional films are just one entry into an expansive genre of animal-as-weapon

10 US Naval Research Laboratory, “Project ORCON: The Use of Pigeons to Guide Missiles,” in *Report of NRL Progress* (Washington, DC: US Naval Research Laboratory, 1959).

11 Harun Farocki, “Phantom Images,” *Public*, no. 29 (January 1, 2004): 17.

12 Skinner, “Pigeons in a Pelican,” 31.

13 Correspondence with John H. Hill, October 12, 1948, box 8, Burrhus Frederic Skinner Papers, Harvard University Archives.

films, including those that feature dogs, orcas, and rhesus monkeys.¹⁴ The Harvard Film Archives contains two versions of the multiple films produced by Skinner, one a completed film titled *PROJECT ORCON: ORganic CONtrol*, and another, incomplete version. In this article, I focus on the completed film, which I believe to be the latest iteration made to promote Project ORCON. It is unclear whether other, previous versions exist and have yet to be found or whether they were taken apart to create each new version. From the surviving examples, it appears that these promotional films were used to dramatically depict the pigeons as reliable and controllable tools. Their imagery presented the birds surrounded by cutting-edge technology, rapidly and competently responding to a dynamic array of changing stimuli. These promotional films played a pivotal rhetorical role in convincing government and private sponsors to back the project. Skinner wrote that one demonstration film was shown “so often that it was completely worn out—but to good effect for support was eventually found for a thorough investigation.”¹⁵ His descriptions of the films’ effectiveness contrast starkly with his accounts of the live presentation of the pigeons’ work, of which Skinner wrote, “The spectacle of a living pigeon carrying out its assignment, no matter how beautifully, simply reminded the committee of how utterly fantastic our proposal was.”¹⁶ Here the moving image performed an essentially symbolic function, primarily concerned with shaping a highly controlled, efficient, and effective image of the weaponized animal bodies.

Film was thus used to transform the pigeon’s behavior both materially and symbolically. Nicole Shukin’s concept of rendering, and its deployment in producing, what she calls “animal capital,” provides a useful theoretical framework for connecting the rhetorical and the functional aspects of the moving image in Project Pigeon.¹⁷ Broadly speaking, animal capital refers to the incorporation of animal life into capitalist systems of currency and exchange. As Shukin writes, “‘Animal capital’ simultaneously notates the semiotic currency of animal signs *and* the carnal traffic in animal substances.”¹⁸ In the history of this processing of animal life, “rendering” takes on a double meaning. On the one hand, it references a history of animal recycling, the process of breaking down animal bodies for the production of a vast array of products, such as meat, glue, and leather.¹⁹ On the other hand, “rendering,” especially recently, often refers to the process of producing a rendition, to the making of an image. Animal capital is the product of this double rendering, where animal bodies are processed into products and consumer goods while their imagery is simultaneously consumed as symbols.

Shukin tracks this history of rendition through the interlocking development of the slaughterhouse and motion studies. She uses the concept of animal capital to examine

14 Heidi Holmstrom, “From War Dogs to Remote Controlled Monkeys: Animals in the Military,” *The Unwritten Record* (blog), June 23, 2015, <https://unwritten-record.blogs.archives.gov/2015/06/23/from-war-dogs-to-remote-controlled-monkeys-animals-in-the-military>.

15 Skinner, “Pigeons in a Pelican,” 35.

16 Skinner, 34.

17 Nicole Shukin, *Animal Capital: Rendering Life in Biopolitical Times*, *Posthumanities* 6 (Minneapolis: University of Minnesota Press, 2009).

18 Shukin, 7.

19 Shukin, 60.

the industrial transformation of organic-inorganic relations in the late nineteenth and early twentieth centuries. In her account, the slaughterhouse is an originary site for the systematic breakdown of bodies in industrial capitalism. Slaughterhouses were the infamous inspiration for Henry Ford's assembly line—although he flipped the process from disassembly to construction—as they literally transformed bodies into an assemblage of component parts. In the slaughterhouse, human labor and animal bodies are both integrated into industrial machinery for the purpose of producing goods. Shukin and others trace the desire to further this process of integrating living bodies into mechanical systems to the initial motion studies of Étienne-Jules Marey and Eadweard Muybridge.²⁰ In those cases, the movements, rather than the corpses, of animals were disassembled and mechanized. The images produced by chronophotography were a new kind of product. They were useful technologies, way stations in the optimization of the animal machine. As Marey wrote in *Animal Mechanism: A Treatise on Terrestrial and Aërial Locomotion*, the 1874 monograph that would inspire Muybridge's chronophotographs: "The comparison of animals with machines is not only legitimate, it is also extremely useful from different points of view. It furnishes a valuable means of making the mechanical phenomena which occur in living beings understood."²¹ The mechanics of flight were contained in the images of a bird's wings, the physics of exertion in the leap of a cat, the dynamics of locomotion in the gait of a horse. Taylorism's integration of chronophotography into the management of the production line completed the circuit of the industrial factory and motion studies, as workers were filmed in order to more efficiently integrate their movements with the demands of the assembly line.²² Within the posthuman logics of animal capital that Shukin postulates, it was only a matter of time before what was done to animals spread to the treatment of humans.²³

Shukin focuses on the slaughterhouse and chronophotography as corresponding sites out of which arose a particular brand of biopolitics and biomechanics in industrial production in the late nineteenth century. But to understand Project Pigeon, we need to study a different history of rendering, one produced by an alternate circuit focused on the science of perception. This is because Skinner's research emerged more directly from the intersection of two different genealogies, those of the camera obscura and the battlefield. The histories of optics and warfare render animals differently than histories of the factory. The animal-as-weapon that is produced from this knotted history is mindless yet complex and fundamentally rational in its operations. Like Marey's chronophotographs, the rendering of the war animals through optical technology was part of a history of harnessing nature's capacities, which in this context was directed at military rather than industrial goals.

20 John Ott, "Iron Horses: Leland Stanford, Eadweard Muybridge, and the Industrialised Eye," *Oxford Art Journal* 28, no. 3 (January 1, 2005): 409–428.

21 Étienne-Jules Marey, *Animal Mechanism: A Treatise on Terrestrial and Aërial Locomotion* (New York: BiblioLife, 2009), 1.

22 Two essential texts on the reimagining of the human body as a machine: Mark Seltzer, *Bodies and Machines* (New York: Routledge, Chapman and Hall, 1992); Anson Rabinbach, *The Human Motor: Energy, Fatigue, and the Origins of Modernity* (Berkeley: University of California Press, 1992).

23 Shukin, *Animal Capital*, 11.

Weaponizing the Nonhuman Sensorium. Well before the advent of Project Pigeon, or capitalism, for that matter, humans and animals were being rendered into useful bodies on the battlefield, consumed not by the industrial factory but by the war machine.²⁴ Animals offer fantastic nonhuman powers for waging war. For centuries, horses, elephants, dogs, and other animals fought alongside human soldiers under almost every conceivable circumstance. These battles often required a deep synchronicity and companionship between humans and animals, creating what David Gary Shaw calls “a unity, a temporary but socially significant fusion of sensible things.”²⁵ At times, these assemblages of human and animal produced deep emotional bonds, with war animals depicted as heroic companions and fellow soldiers. Animals were given medals and funerals, recognized as essential players in the theater of war.²⁶ While they were a means to an end—heightening human senses, providing speed and endurance, elevating commanders above the fray—and were frequently treated as such (often being indiscriminately exterminated or even eaten after their tour of duty), war animals were generally not described in military literature or propaganda as disposable tools in the way that a rifle or a cannon was.²⁷ These charismatic companions ranked as members of the troop or battalion, worthy of recognition and praise.²⁸ Pigeons participated in this history, especially during World War I, when homing pigeons were extensively used to coordinate attacks and relay crucial information.²⁹ As small, speedy, and agile messengers, pigeons enabled commanders to communicate across enemy lines. Additionally, they were known to persist in their missions even when injured, leading to heroic, popular stories of individual pigeons who flew multiple messages even after being wounded. One particularly heroic pigeon, Cher Ami, was stuffed and preserved in the United States National Museum, commemorated by one author as “A Feathered Hero of the World War.”³⁰

24 Alastair Harden, *Animals in the Classical World: Ethical Perspectives from Greek and Roman Texts* (New York: Palgrave Macmillan, 2013); Jilly Cooper, *Animals in War* (London: Corgi, 2000); Steven Johnston, “Animals in War: Commemoration, Patriotism, Death,” *Political Research Quarterly* 65, no. 2 (2012): 359–371; Evelyn Le Chêne, *Silent Heroes: The Bravery and Devotion of Animals in War* (London: Souvenir Press, 2010); Susan Bulanda, *Soldiers in Fur and Feathers the Animals That Served in World War I* (Crawford, CO: Alpine Publications, 2013); Juliet Gardiner and Imperial War Museum (Great Britain), *The Animals’ War: Animals in Wartime from the First World War to the Present Day* (London: Portrait, 2006); “War Culture—Animals of War,” *Military History Monthly* (blog), July 12, 2012, <https://www.military-history.org/articles/war-culture-animals-of-war.htm>; Ryan Hediger, *Animals and War: Studies of Europe and North America* (Leiden, The Netherlands: Brill, 2013).

25 David Gary Shaw, “The Torturer’s Horse: Agency and Animals in History,” *History & Theory* 52, no. 4 (December 2013): 161.

26 Margo DeMello, ed., *Mourning Animals: Rituals and Practices Surrounding Animal Death* (East Lansing: Michigan State University Press, 2016).

27 Elizabeth A. Lawrence, “Animals in War: History and Implications for the Future,” *Anthrozoös* 4, no. 3 (September 1, 1991): 146–147, <https://doi.org/10.2752/089279391787057189>.

28 Hilda Kean, “Animals and War Memorials: Different Approaches to Commemorating the Human-Animal Relationship,” in *Animals and War: Studies of Europe and North America*, ed. Ryan Hediger (Leiden, The Netherlands: Brill, 2012), 237–262.

29 Bulanda, *Soldiers in Fur and Feathers*, 51.

30 Frederick C. Lincoln, “The Military Use of the Homing Pigeon,” *Wilson Bulletin* 39, no. 2 (1927): 67–74.

At the same time, a very different discourse of war animals existed alongside those that at least nominally recognized their contributions as companion species. Animals were just as often used as a kind of expendable technology, capable of killing and being killed in ways that did not risk human soldiers and that were often more effective than traditional weaponry. Stretching back to the Middle Ages, practitioners of biowarfare developed a cruelly efficient use of nonhuman life as killing machines to transform, blight, and infest massive swaths of land.³¹ The germ, the virus, and the swarm were staples in the arsenal of colonialism, where agricultural, ecological, and medical systems were all upended.³² Here, too, pigeons were present. Donna Haraway describes the pigeon as a “creature of empire,” as animals “who went with European colonists and conquerors all over the world,” and were “infamous for ecological damage and biosocial upheaval.”³³ Like rats and crows, pigeons thrived in urban settings, proliferating along with industrialization as signs of the transformation of territory and landscape. By the 1940s, pigeons had also become disposable in a new way as vermin. They were “rats with wings,” symbols of urban blight and decay.³⁴ In this context, pigeons were occasionally associated with other perceived social threats, such as immigrants and the homeless.³⁵ They were a scourge to be eradicated. City officials often ordered the extermination of pigeons in large numbers, and in popular news and magazines they were connected to disease and infestation. In the press and popular imagination, pigeons were not individualized but rather a swarm or pestilence, a threat to public health and sanitation.

Skinner’s Project Pigeon emerges from this history, but it was also particular to its scientific and cultural context, in which the senses and behaviors of animals were being increasingly described through mechanized language. Skinner connected his project to the history of deploying animals’ sensory capacities to extend human awareness of the battlefield, as in the case of the bomb-sniffing dog.³⁶ Like the dog’s sense of smell, the pigeon’s capacity for sight was weaponized for the purposes of seeing from the air. However, Skinner also crucially claimed that Project Pigeon was not about expanding human sensory capability, as pigeon senses do not surpass our own.³⁷ What made

31 Ryan Heider, “Animals and War: Introduction,” in *Animals and War: Studies of Europe and North America*, ed. Ryan Hediger (Leiden, The Netherlands: Brill, 2013), 7.

32 Jared M. Diamond, *Guns, Germs, and Steel: The Fates of Human Societies*, 20th anniversary ed. (New York: W. W. Norton, 2017); Paul Kelton, *Cherokee Medicine, Colonial Germs: An Indigenous Nation’s Fight against Smallpox, 1518–1824* (Norman: University of Oklahoma Press, 2015); Alfred W. Crosby, *Ecological Imperialism: The Biological Expansion of Europe, 900–1900*, 2nd ed. (Cambridge: Cambridge University Press, 2004).

33 Donna Jeanne Haraway, *Staying with the Trouble: Making Kin in the Chthulucene* (Durham, NC: Duke University Press, 2016), 15.

34 Colin Jerolmack, “How Pigeons Became Rats: The Cultural-Spatial Logic of Problem Animals,” *Social Problems* 55, no. 1 (2008): 72–94.

35 Gary Alan Fine and Lazaros Christoforides, “Dirty Birds, Filthy Immigrants, and the English Sparrow War: Metaphorical Linkage in Constructing Social Problems,” *Symbolic Interaction* 14, no. 4 (1991): 391.

36 Skinner, “Pigeons in a Pelican,” 28.

37 Skinner, 28. Skinner writes: “The lower organism is not used because it is more sensitive than man—after all, the kamikaze did very well—but because it is readily expendable.”

the pigeon so valuable in this case was the fact that it was “readily expendable.”³⁸ The pigeon’s perceived disposability allowed it to act as a component within a self-destructing system, seeing and responding up until the last explosive moment, connecting its sensory input to the movements of the bomb throughout the entire flight. From this perspective, pigeons could be ideal artillery, aided by their capacity to learn the complex behavior necessary for being installed into the complicated machinery of the bomb. These birds allowed Project Pigeon to create a bomb that could respond to its environment in real time like a living thing but also be indiscriminately destroyed as an object.

Paired with the disposable pigeon agents, the camera obscura was a crucial integrating device in the development of Skinner’s bomb. It tied the behavior of the pigeon-as-subject to the movements of the living bomb. In fact, the bomb itself was named after a living organism; it was called the “pelican bomb” for its beaklike nose.³⁹ Understanding the role of the camera obscuras in conditioning—or rendering—the behavior of the pigeon within the bomb is essential for understanding the weapon’s function. As Jonathan Crary outlines in *Techniques of the Observer: On Vision and Modernity in the Nineteenth Century*, camera obscuras have a long philosophical history of envisioning, materializing, and conditioning forms of human consciousness.⁴⁰ Skinner’s radical repurposing of this optical tool illustrates his broader vision of warfare fought by distributed autonomous organisms rather than a centralized strategic human intelligence. A camera obscura is constructed by installing a convex lens into a pinpoint hole in one wall of a darkened room, causing the light and images hitting the lens to be projected on the opposite wall. Crary argues that the camera obscura was long used as a means of reconciling observers with the world around them, by modeling the interiority of the subject as both an actual space and a potent metaphor. The symbolic rhetoric of Cartesian dualism was rendered material by the camera obscura, manifesting humanity’s exceptional capacity for reason in its architecture.⁴¹ Within the darkened enclosure created by the camera obscura’s walls was the deliberating, rational human who was given space to reflect, organize, and order the forms and functions of the world outside.⁴² The camera obscura offered its users the power to observe, apprehend, and define at a distance.

In many ways, the perspective produced by the pigeon bomb and other instances of aerial warfare was born out of precisely this humanist logic. As Caren Kaplan describes, scholars studying the history of overhead views have created a narrative in which views from above increasingly compound the split in Cartesian dualism between mind and body, further separating humans from their environment by expanding the

38 Skinner, 28.

39 Skinner, 28.

40 Jonathan Crary, *Techniques of the Observer: On Vision and Modernity in the Nineteenth Century* (Cambridge, MA: MIT Press, 1992).

41 René Descartes, *A Discourse on the Method of Correctly Conducting One’s Reason and Seeking Truth in the Sciences* (1637; Oxford: Oxford University Press, 2006).

42 Crary, *Techniques of the Observer*.

reach of their vision and providing a space for interpretive analysis and reflection.⁴³ Paired with colonial and military projects, the airborne perspectives created by such diverse phenomena as hot-air balloons, spy planes, satellites, and drones have all been tied to the growing expansion of imperial power as they extend the line of sight of a controlling Western “magisterial vision.”⁴⁴ T. J. Demos similarly argues that remote-sensing technologies promise “viewers a sense of control over the represented object of their gaze” in which “the dual colonization of nature and representation[,] appear inextricably intertwined.”⁴⁵ But Kaplan also takes pains to point out that this narrative leaves out crucial breaks and nuances, moments in which “the material and the immaterial are never fully resolved and are therefore productive of ways of knowing and being that do not always square, literally and figuratively, with the Cartesian, bounded subject.”⁴⁶ Poor images, tactile or haptic dissonances, engineering restrictions, realities on the ground, and unexpected affective intensities always threaten to intervene in the narrative of a smooth and seamless increase in human perception and power.

Following Kaplan, we can read Skinner’s Project Pigeon as one of these moments, in that the power promised by the pigeon bomb was predicated on handing over the deadly capacity to perceive and master to a nonhuman consciousness rather than on the expansion of human vision. By placing pigeons in the darkened enclosure of the camera obscura, Skinner hopelessly scrambled the binaries between human and nature, mind and matter, reflection and action, that were represented in more conventional and bourgeois iterations of the camera obscura as well as in later uses of technologies of aerial surveillance, where these abstractions were amplified. Suggesting an alternative historical narrative wherein the Enlightenment march toward total power and abstraction failed, Project Pigeon was the product of several major shifts in thinking and technology from the end of the eighteenth century to World War II. These shifts include changes in optical technology that undermined the distinction between mind and body, developments in zoology that emphasized the connection between media and nonhuman experiences of the world, and the increasing weaponization of optical media to gain an edge on the battlefield.

Shifting Perspectives on Warfare. The experience of rational remove offered to users of the camera obscura was not retained in subsequent visual technologies. As Crary argues, later optical devices, such as the thaumatrope and the zoetrope, effectively located the operations of vision in the physiological processes of the body. For instance, the zoetrope—a cylindrical drum whose inside was lined with successive images, which seem to move when spun and viewed through a series of open slots on the side—clearly displayed the imperfect functioning of the eye, as the viewer could switch back and forth from moving illusion when looking through the slots to incoherent blur when looking over the top of the zoetrope. The marvelous illusions

43 Caren Kaplan, *Aerial Aftermaths: Wartime from Above* (Durham, NC: Duke University Press, 2017), 13–14.

44 Kaplan, 22.

45 T. J. Demos, *Against the Anthropocene: Visual Culture and Environment Today* (Berlin: Sternberg Press, 2017), 28.

46 Kaplan, *Aerial Aftermaths*, 22.

that nineteenth-century media created were premised on a precise manipulation of the senses rather than on the relay of a real world of material objects; in this manner, they effectively disengaged sight from direct access to truth.⁴⁷ The zoetrope was not a metaphor for human rationality and its apprehension of the world as the camera obscura had been but rather an illustration of our capacity to be misled by our senses. Optical media became more and more corporeal, increasingly associating human experiences of the world with the arrangement and often faulty functioning of our sensory apparatus.

At the same time, the particularly nonhuman functions of optical media were also being extolled by those studying animals. The transportive effects of these devices suggested access to other sensoriums beyond the human, other optical truths that might be just as revealing as our own. A parallel interest in the perceptions of animals coming out of ethology in the late nineteenth and early twentieth centuries was also leading to the conclusion that the world experienced by humans was only one of many possibilities. The German ethologist Jakob von Uexküll developed the concept of the *umwelt* to describe the differing perceptual worlds inhabited by human and non-human animals with different sensory capacities. As Jussi Parikka describes, Uexküll “can be thought to show the crumbling of human apperception via the potentially infinite number of perceptual worlds existing in animals—with the world of perceptions too small or too large to comprehend from the human perspective.”⁴⁸ Uexküll described this work as inspired by films made to study animals.⁴⁹ He recounted the experiments of one German researcher who found that certain strains of fish only respond to images projected at thirty frames per second, leading Uexküll to conclude “all processes of motion appear more slowly in their environment, as in slow motion.”⁵⁰ The study of animal perceptions and of moving-image technology both seemed to demonstrate how life could inhabit the world in radically different ways. Unlike the pairing of the Cartesian subject with the camera obscura, Uexküll coupled animals and cinema, with their shared capacity to suggest startlingly different perspectival positions, in order to advocate the exploration of alien perspectives outside of the human.

By the 1940s, optical media had dramatically altered the epistemology of warfare. As Paul Virilio explains in *War and Cinema: The Logistics of Perception*, in the early decades of the twentieth century combat was increasingly fought through the aid of photographic and moving-image technology.⁵¹ Beginning with World War I, battles “depended upon the *regulation of points of view*—that is, on a definition of the battle image in which the cavalry’s perspective suddenly lost out to the perpendicular vision of the reconnaissance aircraft.”⁵² Less and less emphasis was placed on the revelatory

47 Crary, *Techniques of the Observer*, 106.

48 Parikka, *Insect Media*, 64.

49 Jakob von Uexküll, *A Foray into the Worlds of Animals and Humans: With “A Theory of Meaning”* (Minneapolis: University of Minnesota Press, 2010), 70–71.

50 Uexküll, 71.

51 Paul Virilio, *War and Cinema: The Logistics of Perception* (New York: Verso, 1989).

52 Virilio, 73.

capacities of “seeing with one’s own eyes,” and more and more battles were conceived as a panoptic assemblage of nonhuman points of view in combat. Military forces paired the expansive application of surveillance in war with ever-more-powerful artillery, creating a scenario in which one needed only to see the enemy to kill him. As Virilio observes, “The idea of war as fundamentally a game of hide-and-seek with the enemy was proved to the point of absurdity in those First World War earthworks where millions of men were entrenched and interred for four long years.”⁵³ Just as science increasingly offloaded the act of observation onto mechanical means, so, too, the military developed its own “mechanical objectivity,” except with the twist that observation and attack were densely intertwined.⁵⁴ Here the exploratory theories of nonhuman perception were weaponized. For many military strategists, seeing, identifying, and visualizing increasingly became synonymous with monitoring, targeting, and killing. Operationalizing modes of vision beyond those of the individual human combatant reinforced these new associations. The war machine described by Virilio was (and remains) intent on amassing more and more mechanical *umwelts* that could be connected to new methods of reconnaissance and attack.

Therefore, by the time Skinner began his project, military uses of optical media and nonhuman perspectives had been largely intertwined and operationalized for combat. Even in World War I, pigeons were well integrated into this framework. A conspicuous example is Dr. Julius Neubronner’s miniature pigeon camera.⁵⁵ Created in 1903, the pigeon camera was light enough to be carried by a flying pigeon and would automatically snap photographs through a time-released shutter. The German military adopted the pigeon camera as means of surveillance, and a 1916 article in *Popular Science Monthly* reported several such pigeons being shot down by Allied forces.⁵⁶ The article, titled “The Pigeon Spy and His Work in War,” begins with an eloquent appraisal of the pigeon camera’s bizarre mixture of old and new: “It is a strange medley, the air-ship, the last and most daring invention of man’s brain, rising in the early dawn to search out and photograph the foe’s movements, and the graceful pigeon, so frequently mentioned in the stories of early days, soaring, perhaps at the same moment, to act as an aerial scout.”⁵⁷ During World War I, pigeons were part of a loose network connecting intelligence gathering with battalions on the ground, stringing together observation and attack. In particular, pigeons were used to coordinate aerial spy craft. Susan Bulanda notes that “pigeons could be released from aircraft going 100 miles per hour and from heights of up to 6,000 feet.”⁵⁸ Pigeons were used extensively to communicate coordinates to headquarters, an important channel of

53 Virilio, 69.

54 Here I refer to Daston and Galison’s (rather than Benjamin’s) use of the term “mechanical objectivity” to describe the use of visual, auditory, and graphic recording devices in the lab as a replacement for the senses of the human scientist. Lorraine Daston and Peter Galison, *Objectivity* (Cambridge, MA: Zone Books, 2007).

55 “Object Lessons: Dr. Julius Neubronner’s Miniature Pigeon Camera 1903,” *Aperture*, no. 212 (2013): 128.

56 Bonnier Corp., “The Pigeon Spy and His Work in War,” *Popular Science Monthly* 88, no. 1 (January 1916): 30–31.

57 Bonnier Corp., 30.

58 Bulanda, *Soldiers in Fur and Feathers*, 57.

communication for directing the various points of view that were beginning to define the war.⁵⁹

By World War II, military pigeons were thought to be obsolete, with other newly developed means of communication having taken over their functions. In the 1940s proponents of expanding the use of radio and the newly invented radar envisioned these technologies as rapidly connecting bombers to a centralized headquarters and allowing pilots to “fly blind” regardless of the time of day or the weather conditions.⁶⁰ Electronic sensors and communications created the metaphor of war as a grand performance in which remote headquarters coordinated and guided all of its various actors. Despite these developments, there were more pigeons deployed during World War II than World War I.⁶¹ Pigeons were still used to communicate with troops who could not carry the rather bulky equipment necessary for radio and radar communication, pointing to an essential difference between the techno-utopian rhetoric and vision for electronic warfare and realities on the ground.

Nonetheless, Skinner worked to transform the pigeon’s role into that of a possible guidance system for deployed missiles. The threads of embodied media, animal ethology, and optical warfare were all present in this repurposing of the pigeon combatant. Skinner explicitly envisioned Project Pigeon as an alternate operating system to the networks of command and control created by radar and radio.⁶² As Skinner acknowledged, homing missiles that could be guided through radio were also being developed in 1943, but because they existed within the recognized field of electronic warfare, they were also susceptible to interception and jamming.⁶³ No matter how instantaneous the speed of radio communication between bomb and headquarters, there was always the chance of the enemy intercepting or disabling these communications, of severing the connection between the bomb and its controller. Reacting to these flaws, Skinner posited the pigeon as a system of response that was incorruptible in its proximity to the facts on the ground. As an alternative to the grand maneuvers of a centralized orchestration of war, Project Pigeon promised a flock of bird-brained bombs, alive and responding to their environment. Theoretically, the pigeons would self-organize, just as they did in their flocks, and therefore create their own patterns of attack to fit each circumstance.⁶⁴ It was this possible form of warfare that Skinner had suddenly seen on that fateful train ride in 1940—dramatically opposed to the vision of war as a massive centralized organization of many dispersed pieces.

Animals and Devices. Skinner designed the pigeon bomb as a kind of animal, one given a sensorium through optical media and the ability to respond through the

59 Ernest Harold Baynes and Owen Wister, *Animal Heroes of the Great War* (New York: Macmillan, 1925). Baynes and Wister’s 1925 book provides a florid account of pigeons acting as saviors for wrecked and stranded airplanes by allowing them to communicate their location with the head office.

60 Virilio, *War and Cinema*, 78–79.

61 Cooper, *Animals in War*, 104–105.

62 Skinner, “Pigeons in a Pelican,” 31.

63 Skinner, 31.

64 Skinner, *Shaping of a Behaviorist*, 241.

behavior of the pigeon. The falling bomb would sense the space around it through its three camera obscuras, which functioned as eyes. The pigeons inside the bomb would operate as a kind of nervous system, rerouting outside stimuli to the thrusters, allowing the bomb to respond to shifts in position due to drift, air currents, and moving targets. The pigeon bomb was deeply embedded within its environment through a feedback loop produced by sensing and reacting to shifts in perspective. Most of the pigeons' training was dedicated to wedding the nature of the birds to the spectacle produced by the camera obscura, thereby producing an image that would elicit action. The *umwelt* of the pigeon interfaced with the design of the bomb. Unlike the camera obscuras of the seventeenth century, which promoted a sense of separation from the world by walling off the observer inside a space of repose, the images produced in the Project Pigeon bomb were made to connect to the needs of its animal inhabitants. The hungry pigeons were conditioned through days of training to constantly peck certain shapes on the screens in front of them. By rewarding the pigeons with food, Skinner and his peers created an image that elicited action on the part of the observing animal in order to satisfy its basic demands for survival. By intervening into the *umwelts* of the pigeons, Skinner and his research associates could theoretically control the pigeons' responses and tie them to the operational objectives of the bomb.

Toward these ends, the researchers at Project Pigeon, and later at Project ORCON, labored to produce precise simulations of the suicide-bombing missions that the pigeons would be expected to run. While Project ORCON was being conducted, Skinner was researching visual acuity in the pigeon and relaying his research to the ORCON crew, which included testing pigeons' responses as they rapidly moved toward photographs. Skinner consistently pushed the navy's photography division for more clarity and contrast in their images, asking for photographs that were "physically impossible" according to one response.⁶⁵ In attempting to fulfill his request, the photographers had to experiment with new finely ground lenses and large-format cameras. Skinner tested the pigeons' visual acuity in close-up, reproducing the views just before impact. He also simulated the extreme duress that the pigeons would experience in battle by having them practice in different air pressures and by firing a pistol during test runs.⁶⁶

Meanwhile, the staff of Project ORCON created their own elaborate devices for connecting the pecks of the pigeons with their filmed simulations of bombing runs. They invented a relay system whereby each peck created a circuit between a metal headpiece worn by the pigeon and the electrical conducting glass placed in front of the image. The location of each peck was translated as an electrical current registering the distance of that point from the center of the frame. This device was then coupled with a film projector whose projected image would swivel in response to the pigeon's pecks, thereby creating a realistic simulation of what would occur when the pigeons controlled the movement of the bomb. As one report described:

65 Letter from John H. Hill to B. F. Skinner, July 19, 1949, Burrhus Frederic Skinner Papers, Harvard University Archives.

66 "Harvard University Pigeon Project, Final Report," June 15, 1949, Burrhus Frederic Skinner Papers, Harvard University Archives.

The key apparatus here was a small mirror that could be turned right-left and up-down by a servo motor. The motion picture projector beamed the target pictures onto the mirror, which then reflected the images onto the tracking screen. The control loop, embracing the pigeon's beak contact and the conducting glass, provided the signals which determined which way the servo motor would turn the mirror, and thus where the image would appear on the screen.⁶⁷

The result was a film image that changed position in response to where the pigeon struck the screen. These devices simulated the tight feedback loop between organism and environment, allowing for the use and design of the pigeon's responses as a tool. Ultimately, by pairing the on-screen image with the needs of the pigeon's body, and by creating a media device that could respond to its movements, Skinner and his team reimaged the bomb as an optical device that could—with precision—reflect the animal's sensory system, a cyborg apparatus that tethered the life of the pigeon to the working of the bomb.

According to Skinner, the eventual failure of Project Pigeon was not due to a design flaw or incapacity on the part of the pigeons. Instead, it was the result of his own inability to convince the army generals of his dramatically different vision for how war might be fought. Skinner interpreted the final written rejection of Project Pigeon as alluding to the invention of the atomic bomb, a weapon whose sheer power would make the precision of Skinner's missile superfluous.⁶⁸ The world of warfare envisioned with the advent of the atomic bomb, with its emphasis on unilateral human decision making and operational chains of command, was diametrically opposed to Skinner's vision. He had been inspired by watching the coordination of the flock of birds, seeing in their movement a new metaphor for the battlefield. Even though the pigeons' ability to fly ultimately had very little to do with their actual military function, as their wings were strapped to their sides inside the bomb, their flight replaced by the bomb's flight, the perceptual apparatus of the hungry pigeons still guided each bomb, creating a kind of self-synchronizing arsenal of bombs. Skinner's "rendering" (in Shukin's use of the term) not only transformed the pigeon's body into a guidance system but also reimaged a flock of birds as a novel war strategy in which weapons guided themselves and were responsive to their immediate surroundings.

Such a strategy meant relying on the behavior of birds as consistent weapons as well as relying on the work of laboratory psychology to generate replicable scientific findings that could be applied in circumstances of life and death. Orienting animal laboratory research toward the production of weapons based on avian behavior required adapting the terms of psychology to the language of the military-industrial complex. As James Capshew writes in his account of the scientific history of Project Pigeon, Skinner and his researchers were initially unable to convince even the other scientific divisions of the validity of their findings, although they "eventually [learned] to articulate their work in engineering terms, as seen in the use of the metaphor of the

67 US Naval Research Laboratory, "Project ORCON."

68 Skinner, "Pigeons in a Pelican," 34.



Figures 2 and 3. Panels depicting Project Pigeon from a comic strip in the October 11, 1959, edition of the *Toledo Blade*. Copyright *The Blade* 1959.

bird as a machine.”⁶⁹ Both the success and the failure of Project Pigeon hinged on this “rhetorical ploy,” as Capshew calls it, which described the pigeon as a dependable piece of equipment that could be trusted to operate on its own without oversight from a human commander.⁷⁰ As Skinner wrote: “We had begun to realize that a pigeon was more easily controlled than a physical scientist serving on a committee.”⁷¹

Here, again, film was crucial, not in the implementation of Skinner’s model of war but in its promotion—in his ability to sell his idea to the military brass. Here again, Skinner struggled to get Project Pigeon taken seriously.

He described being all but laughed out of conference meetings on several occasions. Indeed, when the project was eventually declassified, it was the butt of many jokes by popular commentators. An example of this scorn can be seen in a 1959 cartoon for the *Toledo Blade* (see Figures 2 and 3) in which a pigeon is depicted dressed as a pilot boasting to another pigeon in a black cocktail dress, “I’m a big missile man from U. of M.” The humor of the joke is premised on the absurdity of the pigeon replacing the human pilot, the incongruity between the prestige and authority of the pilot’s role and the animal body of the pigeon, and the ludicrousness of a pigeon boasting of his flights over a cocktail. Project Pigeon seemed laughable as long as the pigeons were viewed as living substitutes for decisions made by pilots, as we see in a different panel of the same comic that depicts two pigeons inside the bomb debating whether they are heading in the right direction. As devices designed and installed into a machine, pigeons were considered by military funders to be an efficient and cheap means to an end, but as decision-making actors, they became anthropomorphic caricatures of war strategists and bombardiers placing the power of the US Air Force in the hands of animals.

The frequent dismissive responses to Project Pigeon, exemplified by these comic strips, stand in utter contrast to the stark and brutal efficiency depicted in the project’s promotional films. Skinner’s film had “good effect,” as he repeatedly put it, in helping to procure funding at several steps in research and development.⁷² General Mills, the initial sponsor of Skinner’s experiments, and Franklin Taylor, the eventual director of Project ORCON, used footage from Skinner’s film to market the weapon to possible investors in the competitive field of military contracting. This film was one entry into the small subgenre of nontheatrical movies made to promote animal weaponry in the military.⁷³

69 Capshew, “Engineering Behavior.”

70 Capshew, 853.

71 Skinner, “Pigeons in a Pelican,” 31.

72 Skinner, 31, 35.

73 Holmstrom, “From War Dogs to Remote Controlled Monkeys.” Held at the National Archives, *The Use of Dogs in War* (1943), *A Technical Film Report on Project Deep Ops* (1972), and *Paisley Print Task I* (1985) are all good examples of this genre, featuring dogs, orcas, and rhesus monkeys, respectively.

Such films generally depict nonhuman organisms as integral parts of a modern military arsenal: a biodiversity of weapons, each contributing a body and behavior ready and available for every circumstance. Humans are shown on the periphery, installing and operating tech that will direct the movements of the war animal. Within this weaponized menagerie, the human viewers are instructed on the wartime use of animals, apprised of the development of new biotechnology, and encouraged to imagine a battlefield in which animal combatants swim, swarm, and scamper into the fray as proxies for human soldiers.

To that end, the scenes shot to promote Project ORCON picture the bodies of the pigeons as capable of full integration into machine technology. Viewers are embedded in the highly technical space of the behavioral lab where the pigeon is conditioned to follow and peck certain shapes. The film begins with the installation of a headpiece on a pigeon in extreme close-up; the viewer sees how its beak becomes a part of the bombing instrument. The hands of the scientist cup and frame the head of the pigeon as a small metal prong is stuck to its forehead. Held between what appear to be two giant fingers, the pigeon's head is turned side to side, providing both a more complete image of the installed headpiece and a demonstration of the bird's malleability and compliance (Figure 4). The pigeon is subsequently shown performing in a series of simulations that display its ability to target metallic objects that slide on rails behind an open screen. Viewed from behind, the construction of the film emphasizes the typewriter-like rapidity of the pigeon's head as it matches its movements to the automated back-and-forth of the target. Here, the pigeon's animality is depicted as a powerful rapid-response system capable of keeping time with the other moving mechanisms surrounding it.

The final two shots depict the training of the pigeons responding to Project ORCON's filmed simulations. These scenes repeatedly show the pigeon guiding the missile to a series of ships in silhouette (Figure 5). Each time a battleship veers from the center, the pigeon reorients the image by accurately pecking on the ship, shifting the picture's position on the circular screen to reframe the target. Shown in a magnified close-up, the film constructs a bomb's eye view. This is a triumphant image for the power of merging animal with machine, placing viewers with the pigeon inside the bomb as it moves closer and closer to the oncoming ship. Each shot cuts off just before impact, suggesting the final destruction of pigeon, bomb, and boat. The effect of these shots is one of brutal efficiency, in which the pigeon continues to perform its compliant, nearly automatic responses up until the last moment, at which point the image and the bird both disappear. Then, after briefly cutting to black, the film repeats the dive again. Here, the multiplication of similar shots evokes a flock of similar bombs, each inhabited by small, disposable kamikaze pilots who could dispassionately destroy themselves and the enemy with expert accuracy. Revealing fairly little detail of the actual workings of the bomb, the final shots are more iconic than informative, prompting the audience to imagine a new form of bioweaponry and war.

Contemporary Resonances. In her analysis of contemporaneous military research, Katherine Chandler demonstrates how the initial stirrings of modern drone



Figures 4 and 5. Images from film made to promote Project Pigeon. Still captures from a digital file. Courtesy of the B. F. Skinner Foundation.

warfare can be seen in early military experiments into televisually guided missiles.⁷⁴ Throughout the 1930s and 1940s, the technology for remotely controlled assault aircraft was developed and tested. However, many in the military ultimately felt that only a craft manned by a human aviator could reliably be put to use on the battlefield.⁷⁵ The military had the required technology to engage in something like modern drone warfare, and yet, Chandler concludes, it lacked the interpretive framework to see that

74 Katherine Chandler, "American Kamikaze: Television Guided Assault Drones during World War II," in *Life in the Age of Drones*, ed. Caren Kaplan and Lisa Parks (Durham, NC: Duke University Press, 2017), 152–188.

75 Chandler, 99.

potential. A similar claim can be made about Project Pigeon. While Project Pigeon was never realized as a military technology, we now inhabit a future where Skinner's living weapons are no longer entirely a "crackpot idea" and the "natureculture" of the battlefield is increasingly recognized and put to lethal use.⁷⁶ As we grapple to understand this present moment, it is instructive to look back at Skinner's failed project, envisioned while gazing out a train window, to transform a flock of birds into an arsenal. The two-pronged rendering of bodies and images of war animals continues today, as weapons manufacturers deploy animal physiologies and adapt ecological images to military ends. Body armor made from spider silk, remote-controlled insect drones, bomb-detecting harbor seals, mine-detecting dogs, and other animal cyborgs populate our military armory.⁷⁷ Additionally, progress in artificial intelligence, robotics, and genetic engineering has accelerated the use of animal weaponry far beyond anything Skinner could have imagined. Even products of purely mechanical engineering seem to be possessed by the ghosts of animal combatants, as the military increasingly depends on nonhuman proxies such as the BigDog and WildCat combat robots. Again, optical technology is essential for the creation of such weapons. Thanks to the integration of the moving image within nonhuman response systems, we have become accustomed to thinking about our weapons as having a perspective, a point of view, and of being capable of responding to changing events on the ground. Caren Kaplan describes how military intelligence's conception of surveillance technology has recently shifted from "identifying fixed, precise locations to 'situational awareness' in relation to a 'field of motion.'"⁷⁸ Flexible, autonomous, seemingly lifelike weaponry is being developed to respond to the movements of individual combatants as they populate urban and civilian areas rather than to survey the movements of armies. As the chief of the Network Science Division of the Army Research Laboratory, Alexander Kott, recently observed: "A variety of networked intelligent systems—things—will continue to proliferate on the battlefield, where they will operate with varying degrees of autonomy. Intelligent things will not be a rarity but a ubiquitous presence on the future battlefield."⁷⁹ These new developments realize Skinner's radical vision of warfare not only as a contest of solely human soldiers but also as an elaborate network of animals and machines that mobilize nonhuman bodies, movements, and perceptions.

The ethics of this shift have always been questionable. For his part, Skinner, who became a vocal antiwar advocate during the Vietnam War, observed: "The ethical

76 Here I draw from Haraway's term "natureculture," which both acknowledges the powerful separation of nature and culture as philosophical and political categories and simultaneously highlights a history of material interactions that undermine this division. Donna Haraway, *The Companion Species Manifesto: Dogs, People, and Significant Otherness* (Chicago: Prickly Paradigm, 2003), 13.

77 "Mine Hunting Systems of the US Navy Marine Mammal Program," February 8, 2006, https://web.archive.org/web/20060208143241/http://www.spawar.navy.mil:80/sandiego/technology/mammals/mine_hunting.html; Bridget Heos, *Stronger than Steel: Spider Silk DNA and the Quest for Better Bulletproof Vests, Sutures, and Parachute Rope* (Boston: HMH Books for Young Readers, 2013); Peter Shadbolt, "Robo-Wings: Military Drones That Mimic Hawks and Insects," CNN, January 14, 2015, <http://www.cnn.com/2015/01/14/tech/mci-drone-robhawk-robofly>.

78 Kaplan, *Aerial Aftermaths*, 213.

79 Alexander Kott, "Challenges and Characteristics of Intelligent Autonomy for Internet of Battle Things in Highly Adversarial Environments," *arXiv:1803.11256 [Cs]* (Ithaca, NY: Cornell University), March 20, 2018.

question of our right to convert a lower creature into an unwitting hero is a peacetime luxury. There were bigger questions to be answered in the late thirties.”⁸⁰ In the face of the horrors of World War II, Skinner felt that the loss of pigeon lives was a small price to pay in a battle against enemies who had “promised, and eventually accomplished, the greatest mass murder in history.”⁸¹ In many respects, however, the distributed agencies of the battlefield precipitated by Project Pigeon have effectively erased the distinction between peace and wartime, creating an endless sense of urgency about events on an increasingly remote battlefield. As Brian Massumi describes, the preemptive logic of the “war on terror” no longer relies on calls for direct intervention but rather rests “on the wings of a drone.”⁸² Realizing the promise of the pigeon bomb, drone warfare has allowed America to conduct a never-ending war of extrajudicial killings without either a declaration of war by Congress or any real reckoning of American intervention abroad.⁸³ The lingering ethical questions of who is responsible for remote killings and what the effects are of such asymmetrical risk to human life stay with us today. The distributed agency of the pigeon bomb, which so startled the US commanders during World War II, has become a reliable political and strategic tool in contemporary warfare, where the accountability and costs of war have been dispersed to swarming flocks of nonhuman actors. As we come to grips with the fact that there is no “peacetime” to look forward to, no respite when we can pause and debate these approaches, the ethical questions about our newly accepted nonhuman combatants cannot wait. *

80 Skinner, “Pigeons in a Pelican,” 28.

81 Skinner, 28.

82 Brian Massumi, *Ontopower: War, Powers, and the State of Perception* (Durham, NC: Duke University Press, 2015), 18.

83 Massumi, 92.